E-RANK: DIGITAL TAXI RANK MANAGEMENT AND PASSENGER SAFETY SYSTEM

SYSTEM REQUIREMENT SPECIFICATION: PHASE 2

GROUP MEMBERS:

Oarabetse Morata 202406427

Phuti Setati 202435062

Kholofelo Phalakatsela 202306829

Louisa Mdluli 202324412

Siyabonga Ndzobondzobo 202441850

Group Name: PMP Solutions

Due date: 20 April 2026

Github Repo Link: https://github.com/SiyaJNdzobs/NPRT63PMPSOLUTIONS.git

Table of Contents

Introduction:

- This document outlines Phase 2 of the E-RANK Project, building on Phase 1 by translating identified problems into detailed system specifications. Phase 1 highlighted key issues such as lost passenger information, disputes over taxi queues, low visibility for taxi owners, and passenger uncertainties. Phase 2 focuses on defining the processes for using the E-RANK system, analysing functional and non-functional requirements, and providing UML diagrams to detail system behaviour. Additionally, it includes a work breakdown structure and Gantt chart to offer realistic estimates for the system's implementation and roll-out. The major actors remain unchanged from Phase 1, with the Taxi Marshal as the primary controller, Drivers operating the taxis, Taxi Owners overseeing fleet operations, and Passengers seeking reliable information about taxi ranks. This phase ensures the system addresses previous inefficiencies while setting a structured plan for development and deployment.

1.Business Process:

A business process explains how users go about doing things using the software to achieve their goals in the software environment. Different from features, a business process explains the series of activities performed by the user in the software and the response of the system. The following section provides five key business processes that solve the major challenges highlighted in Phase 1.

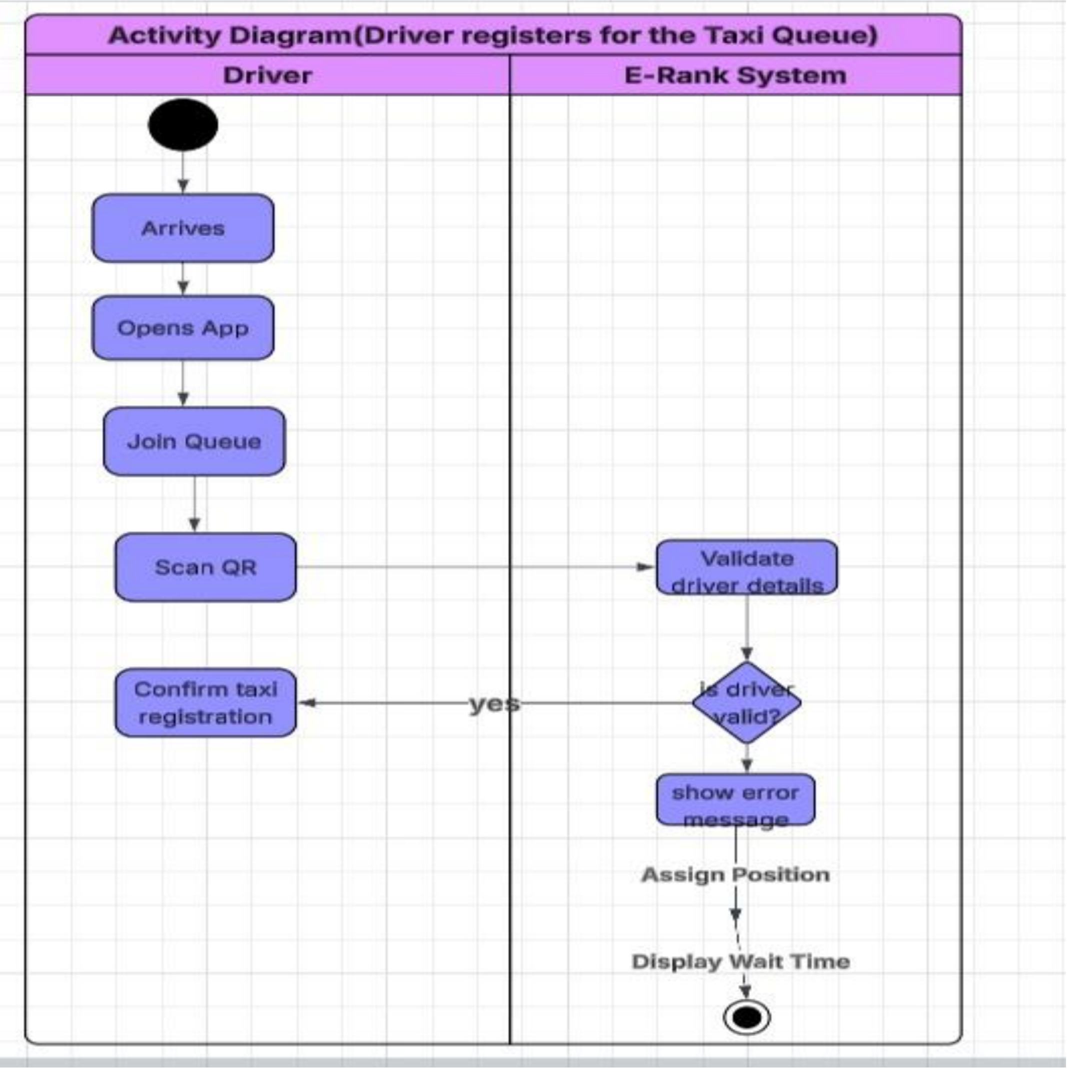
A process is explained in the form of a flow chart and then a summary in a tabular format highlighting the distinction between user and system activities. One person evaluating this process should be able to follow the steps without any confusion.

1.1: Driver Registers for the Taxi Queue

This process handles how drivers register their position in the digital taxi queue upon arriving at the rank

	User Action	**System Response**
Step 1	Driver arrives at taxi rank entrance.	System is ready to accept QR code scans.
Step 2	Driver opens E-RANK mobile app and taps "Join Queue".	System activates device camera for QR scanning.
Step 3	Driver scans the QR code displayed at the rank entrance.	System captures driver ID, taxi registration, and timestamp.
Step 4	Driver confirms taxi registration number on screen.	System validates that the driver is registered and authorised.
Step 5	Driver waits for confirmation.	System adds driver to end of digital queue and assigns queue position number.
Step 6	Driver receives confirmation message.	System displays: "You are position #5. Estimated wait: 12 minutes."

Figure 1.1: Driver Registers for the Taxi Queue:



1.2: Registration of Long-Distance Passenger by the Taxi Marshal

The first phase showed that passengers for long-distance travel have their details recorded in record books. However, these documents are prone to loss or damage. This process takes away such issues since passenger registration is conducted electronically on the boarding point.

In case a passenger wants to use the long-distance taxi service, he or she is supposed to approach the taxi marshal. At this stage, the taxi marshal requests for the passenger's South African ID document and his or her intended destination. Afterward, the marshal types the 13-digit ID number of the individual and adds the destination to the tablet's registration form. The system confirms whether the provided ID number is valid or not according to its format. The marshal then selects an available taxi through the dropdown list of taxis available at the time of registration. After selecting the appropriate taxi, the marshal touches the 'Register Passenger' button on the form. Finally, the system records the data, encrypts it for confidentiality purposes, associates the passenger with a certain taxi and driver, and then stores the information in the cloud.

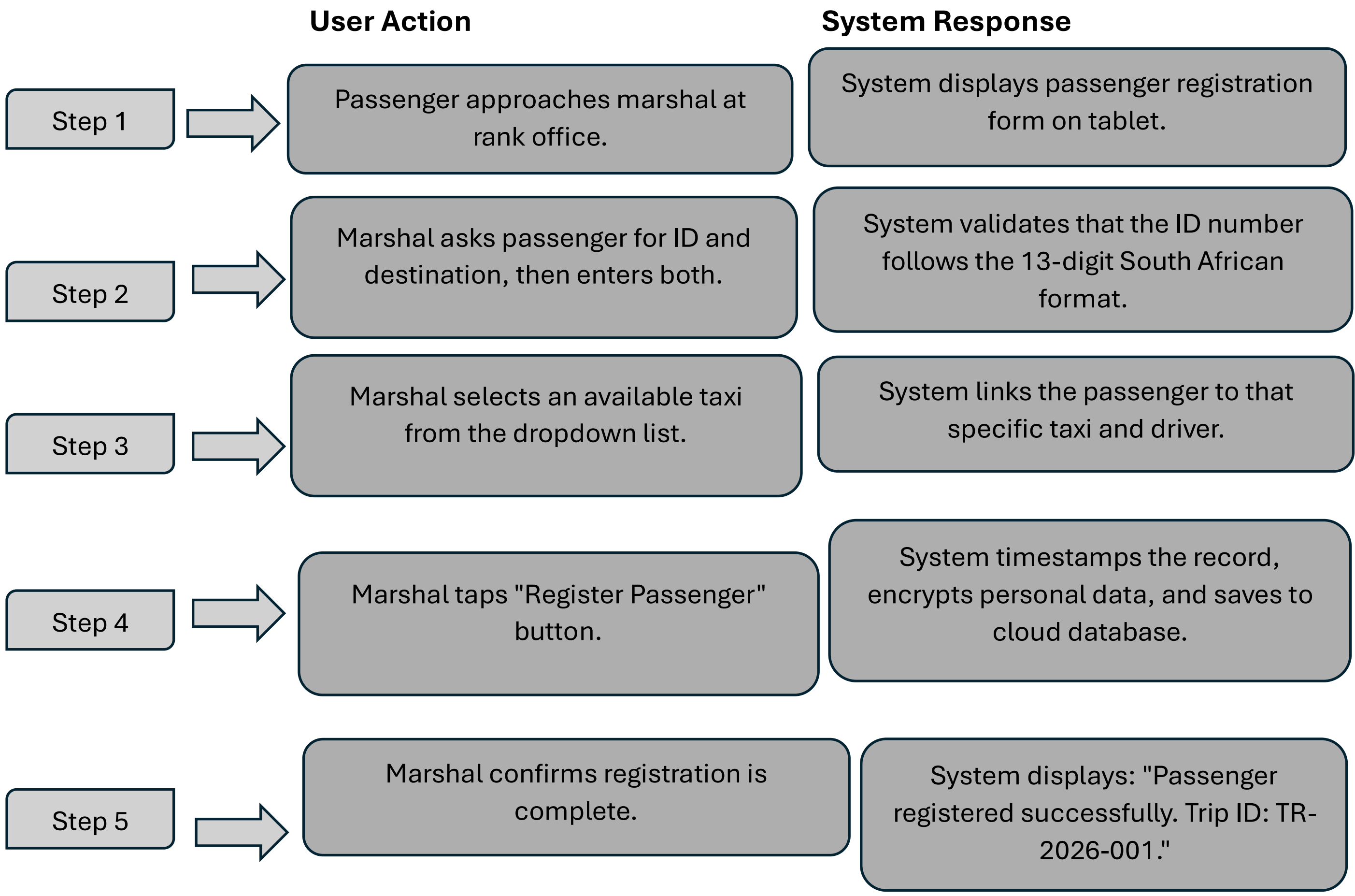
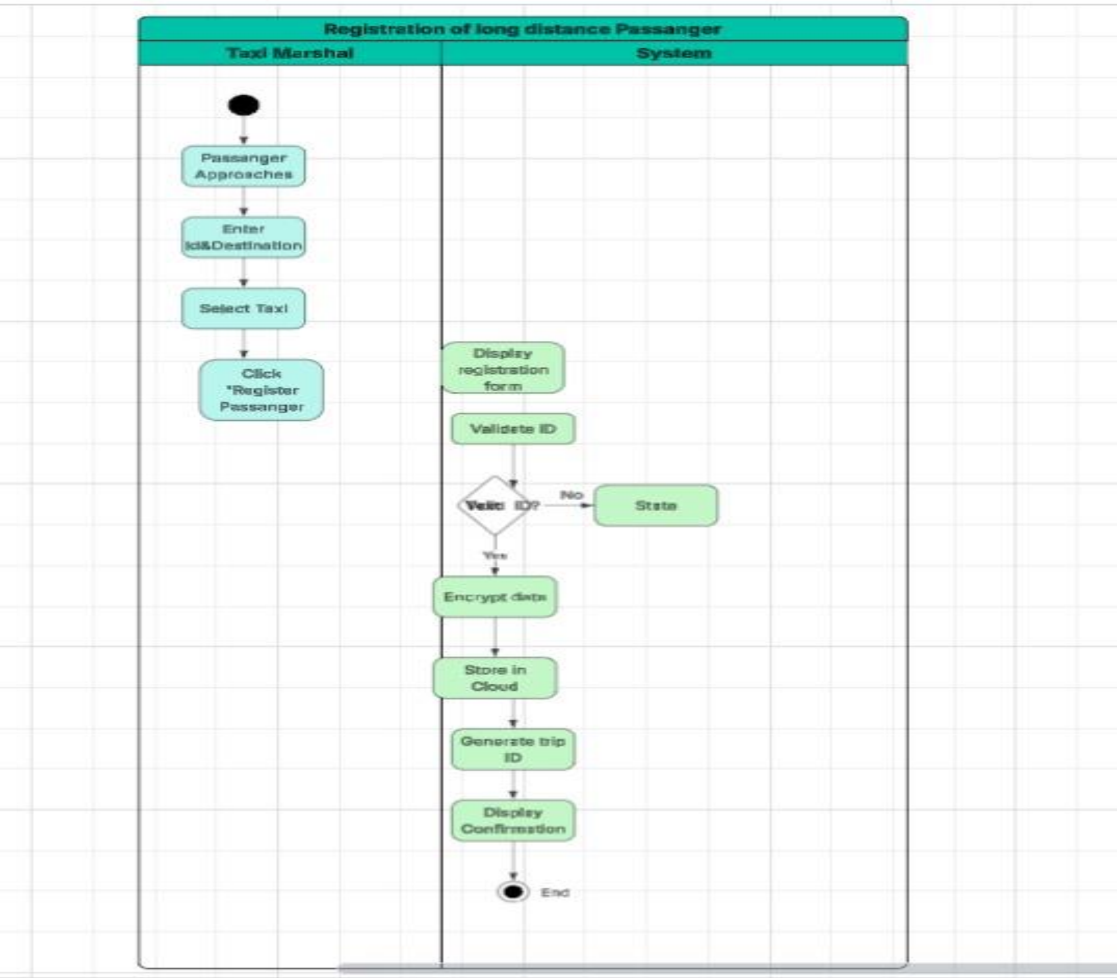

User Action
System Response
Step 1
Passenger approaches marshal at rank office.
System displays passenger registration form on tablet.
Step 2
Marshal asks passenger for ID and destination, then enters both.
System validates that the ID number follows the 13-digit South African format.
Step 3
Marshal selects an available taxi from the dropdown list.
System links the passenger to that specific taxi and driver.
Step 4
Marshal taps "Register Passenger" button.
System timestamps the record, encrypts personal data, and saves to cloud database.
Step 5
Marshal confirms registration is complete.
System displays: "Passenger registered successfully. Trip ID: TR-2026-001."

Figure 1.2: Activity Diagram – Registration of Long-Distance Passenger by the Taxi Marshal:



1.3: Taxi Marshal Manages the Digital Taxi Queue

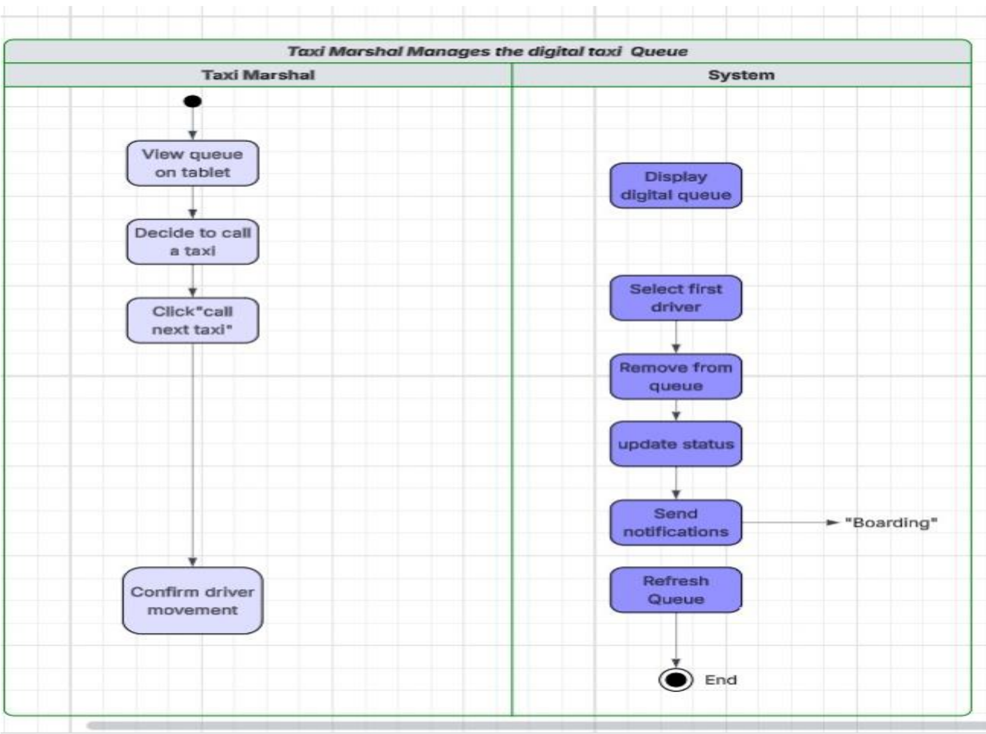
This process describes how the marshal uses the E-RANK system to maintain order and call taxis for boarding. The manual alternative described in Phase 1—using informal queue systems and verbal communication—frequently leads to disputes. The digital system eliminates ambiguity by providing a single, authoritative queue order that all drivers can see.

The marshal begins by looking at the E-RANK tablet, which displays a real-time list of all drivers currently waiting in the queue. Each driver is shown with their name, taxi registration number, and queue position. When the marshal determines that a taxi is needed for boarding—typically because a previous taxi has departed and new passengers are waiting—the marshal taps a "Call Next Taxi" button.

The system automatically selects the driver at the front of the queue and removes that driver from the waiting list. The driver's status is then changed from "Waiting" to "Boarding". Simultaneously, the system sends a push notification to that driver's mobile application, informing them that they have been called to move their taxi to the boarding area. The marshal then informs the driver verbally or through a designated signal to proceed. Once the driver moves, the marshal confirms the action on the tablet, and the system refreshes the queue display so that all remaining drivers shift up one position.

	User Action	System Response
Step 1	Marshal looks at tablet to see waiting taxis.	System displays digital queue with driver names, taxi registrations, and queue positions.
Step 2	Marshal determines a taxi is needed for boarding.	System highlights the driver at the front of the queue.
Step 3	Marshal taps "Call Next Taxi" button.	System removes the front driver from the queue.
Step 4	Marshal confirms the driver selection.	System changes driver status from "Waiting" to "Boarding".
Step 5	Marshal informs driver to move to boarding area.	System sends push notification to driver's app: "You are called to board."
Step 6	Marshal confirms driver has moved.	System refreshes queue display. Remaining drivers shift up one position.

Figure 1.3: Activity Diagram – Taxi Marshal Manages the Digital Taxi Queue:



1.4 Driver Verifies Completion of a Ride

As observed during phase one, taxi owners do not know how many rides their taxis are doing per day. This process helps in monitoring rides by allowing drivers to verify when a ride is over.

The driver first stops at the intended destination and then opens the E-RANK driver app on his/her phone. The app shows a list of all ongoing rides assigned to the driver. The driver chooses the right ride on the list based on information such as the destination and the number of passengers. Then the driver clicks the 'End' button.

The app notes the exact time and deducts the initial time when the taxi left the rank to calculate the total duration of the ride. It then marks the ride as 'completed.' The app automatically adds the fare charged for the ride, which was previously entered by the marshal when registering the passengers, to the daily summary of the taxi owner. Finally, the driver gets a notification with the total fare amount and a report on the ride.

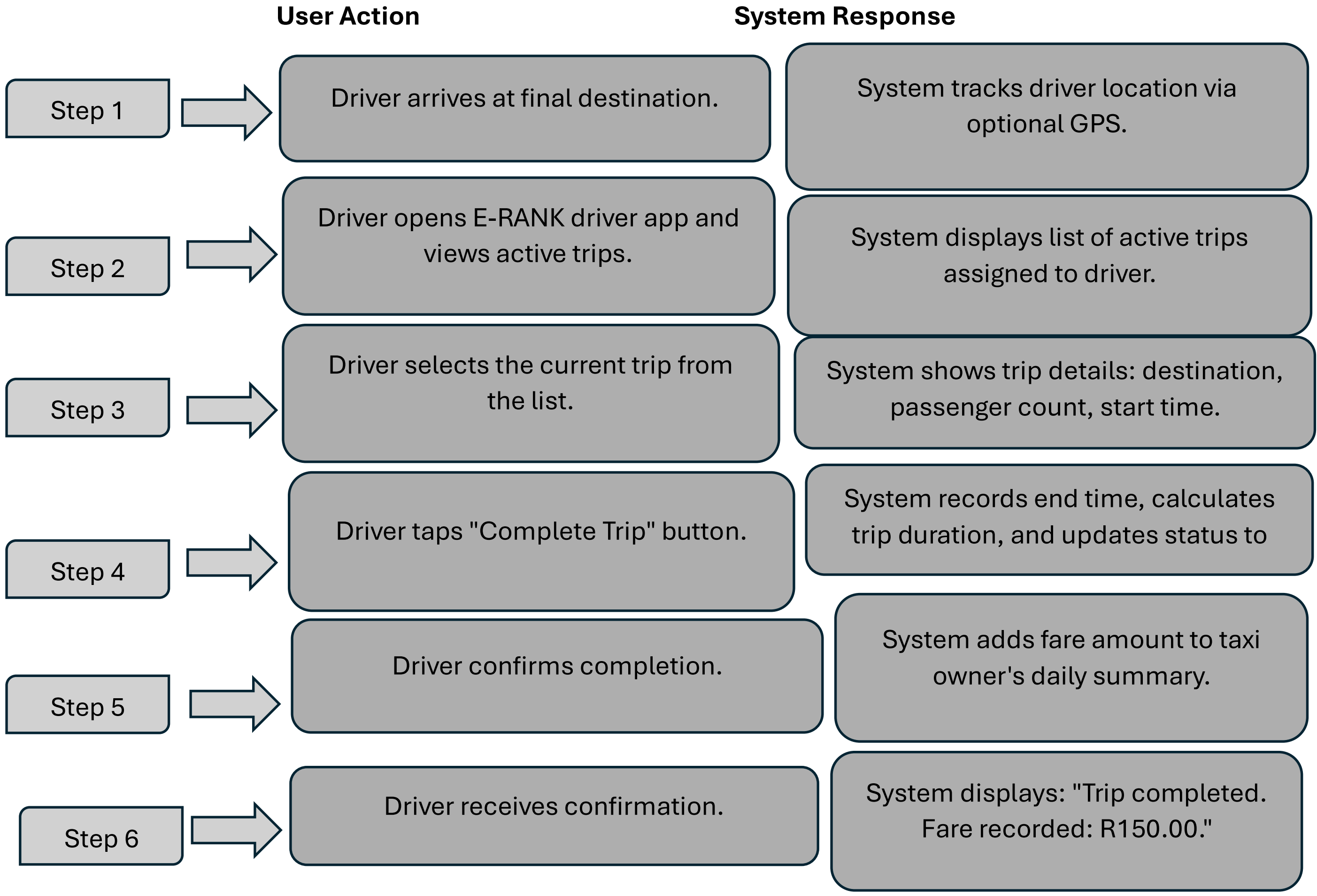
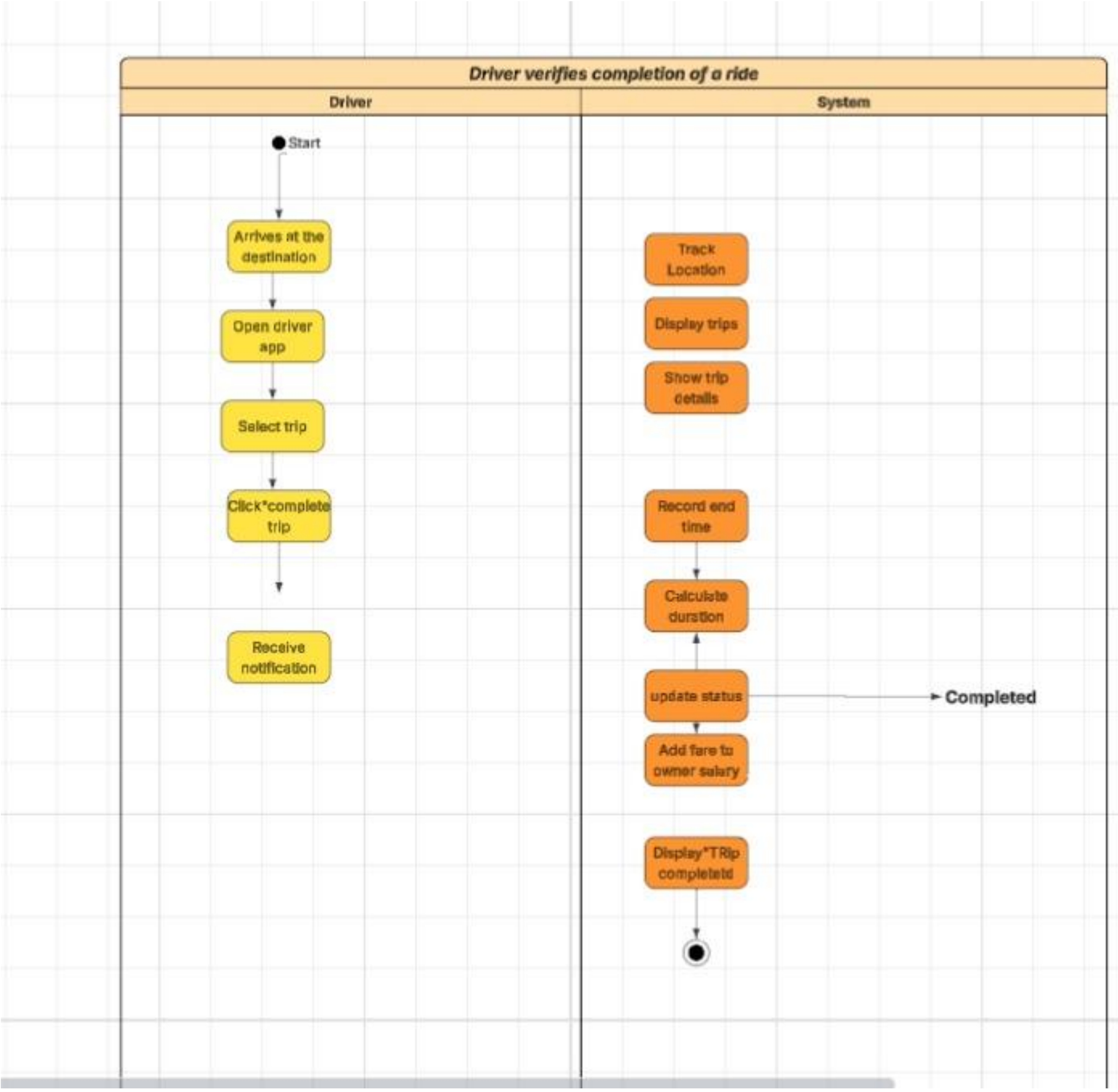


Figure 1.4: Activity Diagram – Driver Verifies Completion of a Ride:



1.5 Owner Monitor Fleet Performance

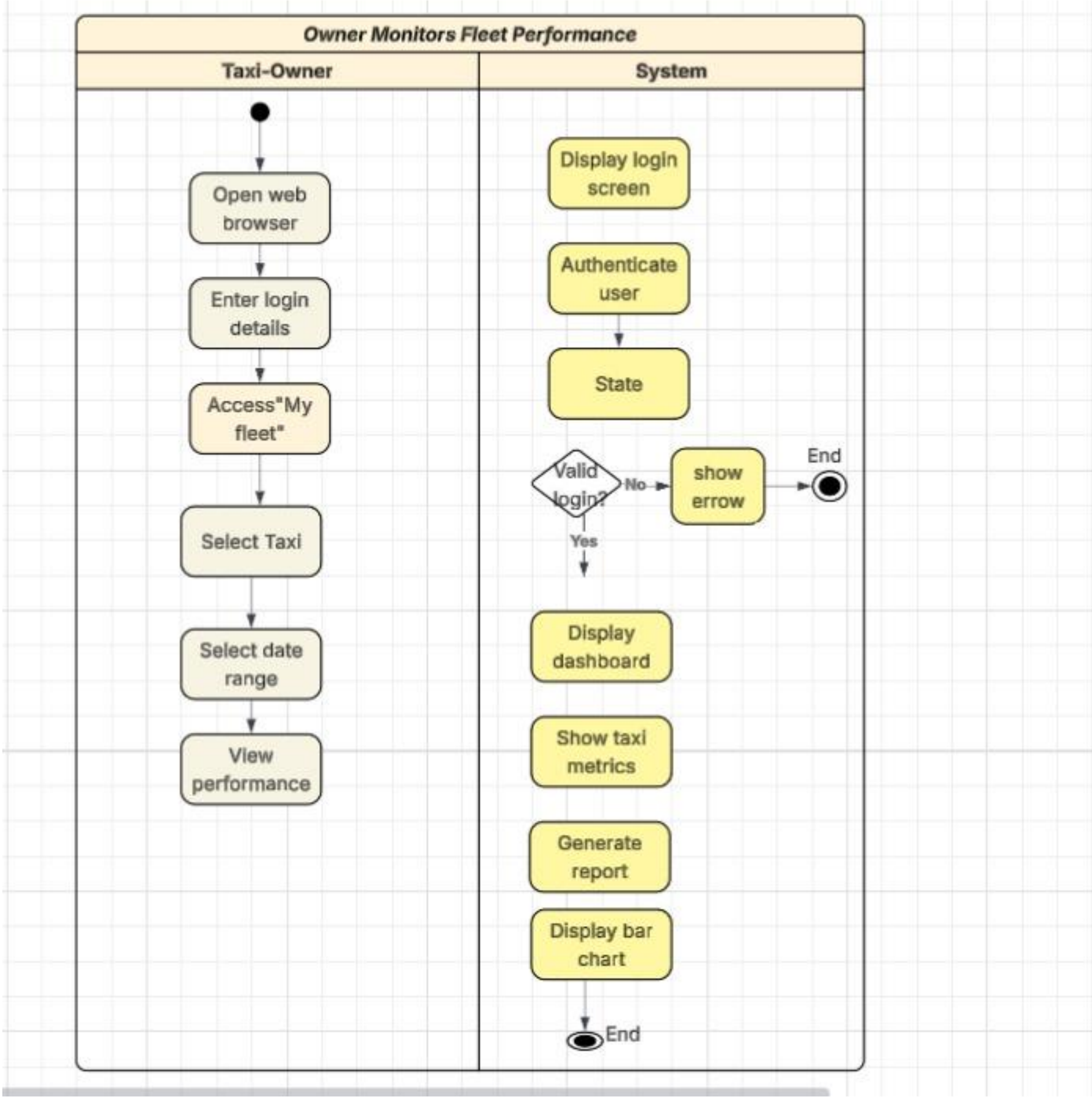
This process aims at addressing the first phase problem that owners lack insight into how their fleet of taxis operates on a day-to-day basis. The E-RANK platform ensures that an owner dashboard is put in place where all trips performed for the owner's taxis can be monitored.

The owner begins the process of monitoring their taxis by opening the web browser on their personal computer and entering the E-RANK owner portal. Next, the owner logs in by providing their username and password. After validation and authentication, the owner gets access to a "My Fleet" dashboard where an overview of all their registered taxis is displayed. This includes the total number of trips made by all taxis, total earnings made through these trips, and the status of each taxi in regard to whether it is "at rank", "en route", and offline".

It will respond by providing performance statistics related to that particular car. These include total trips undertaken today, total fare collected, name of the current driver, and the current status of that car in terms of queues. In case the owner wants to analyse the performance for an extended period, he/she can choose a custom date range like seven days or even a month. In turn, the system will generate a trip report, which comes in form of a basic bar chart indicating trips undertaken daily.

	User Action	System Response
Step 1	Owner opens web browser and navigates to E-RANK owner portal.	System displays login screen.
Step 2	Owner enters username and password.	System authenticates credentials and verifies owner permissions.
Step 3	Owner taps "My Fleet" menu.	System displays high-level summary of all owned taxis: total trips, total fares,
Step 4	Owner selects a specific taxi by license plate.	System displays detailed metrics: trips today, fares collected, current driver, queue status.
Step 5	Owner selects a custom date range (e.g., "Last 7 days").	System generates trip history report and bar chart of trips per day.
Step 6	Owner reviews performance data.	System maintains report for export or printing.

Figure 1.5: Activity Diagram – Owner Monitors Fleet Performance:



3. Functional Requirements

- Functional requirements define what the E-RANK system should achieve. In contrast to non-functional requirements, which represent the quality characteristics of software, functional requirements represent concrete inputs, actions, and outputs. Every functional requirement mentioned in this section provides a solution to the identified problems in Phase 1.

For instance, one of the problems identified in Phase 1 states that "passenger information for long-distance rides is written down in paper-based registration books, which are likely to be misplaced or even destroyed." The corresponding functional requirement (FR-01) requires the system to have a passenger registration function that will digitally store passenger information in a cloud database.

In addition, the problem mentioned in Phase 1 says that "the taxi queue is managed manually, leading to disputes among drivers about queue positions." Functional requirement FR-02 requires the system to manage a digital queue by generating unique queue numbers visible to all participating drivers.

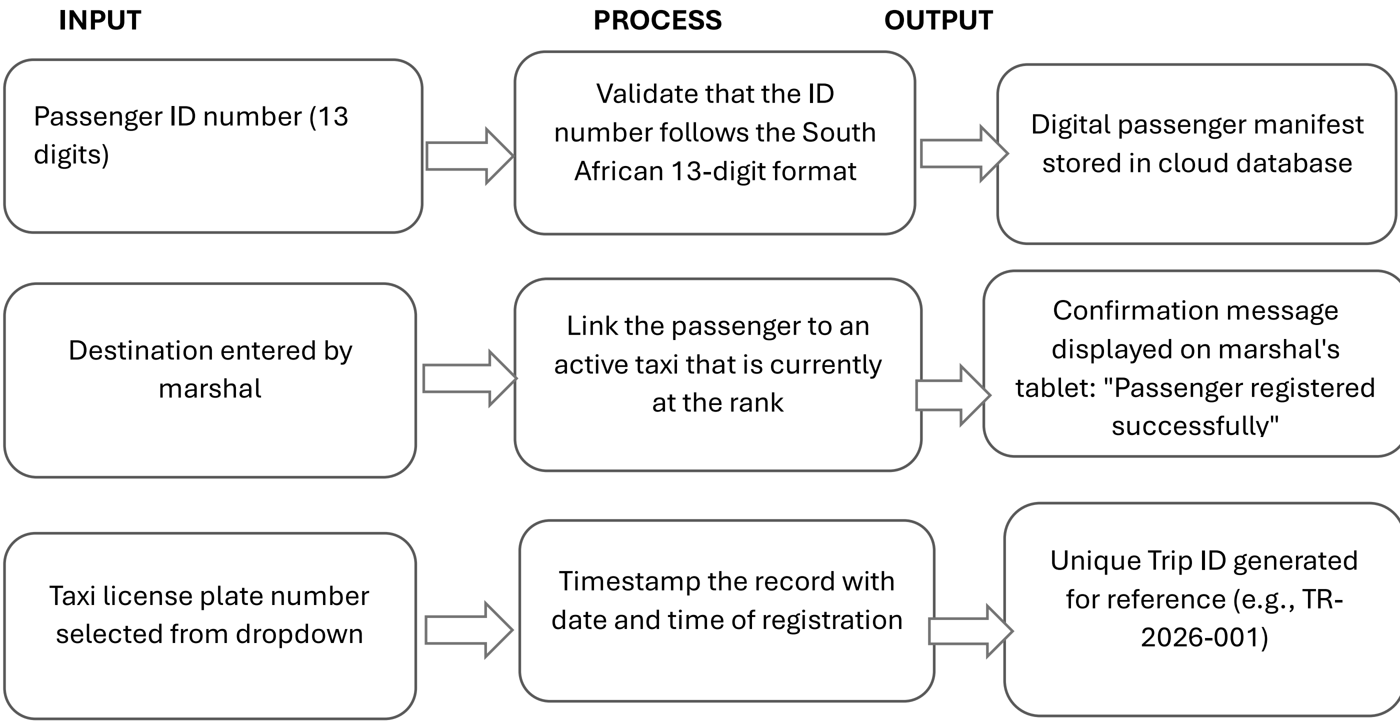
The following table represents six functional requirements. It provides the user who performs the action, necessary inputs, process performed by the system on the inputs, outputs provided to the user, and a concise explanation of how each requirement solves the problem mentioned in Phase 1:

FR-01: Register Passenger for Long-Distance Trip

User: Taxi Marshal

Problem Solved (from Phase 1): Passenger records are lost or damaged when stored in physical books.

Description: The system shall allow the Taxi Marshal to digitally register a passenger who is booking a long-distance trip. This eliminates the need for physical record books and ensures that passenger information is securely stored and easily retrievable.

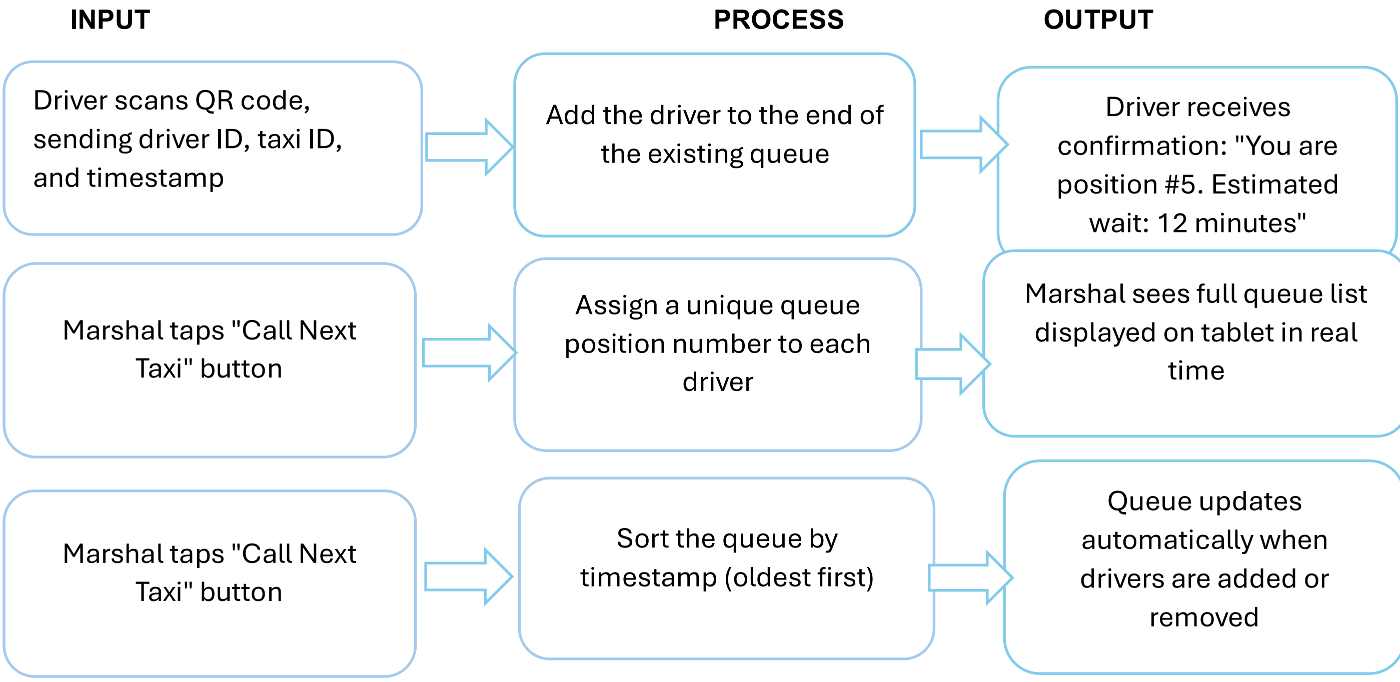


FR-02: Maintain Digital Taxi Queue

User: Taxi Marshal

Problem Solved (from Phase 1): Queue disputes occur between drivers because queue order is not digitally recorded.

Description: The system shall maintain a digital queue of all drivers waiting for passengers at the rank. Each driver receives a unique queue position, and the marshal can view and manage the queue from the tablet. This provides an authoritative record that eliminates disputes.



FR-03: Display Trip Summary for Taxi Owner

User: Taxi Owner

Problem Solved (from Phase 1): Owners have limited visibility into how many trips their vehicles complete each day.

Description: The system shall provide a dashboard for taxi owners to view trip summaries for each of their taxis. The owner can select a specific vehicle and date range to see performance metrics.

INPUT	PROCESS	OUTPUT
Owner selects a taxi ID from dropdown menu	Query the database for all completed trips matching that taxi ID	Report showing total trips completed in the selected period
Owner selects a date range (e.g., "Last 7 days")	Calculate the total number of trips completed	Report showing total fares collected
Owner clicks "Generate Report" button	Calculate the average trip duration	Simple bar chart showing trips per day

FR-04: Display Route and Taxi Availability to Passengers

User: Passenger

Problem Solved (from Phase 1): Passengers experience uncertainty about taxi availability, routes, and operating hours.

Description: The system shall provide a public-facing interface (mobile app or website) where passengers can check which routes are available from a given rank and how long they can expect to wait.

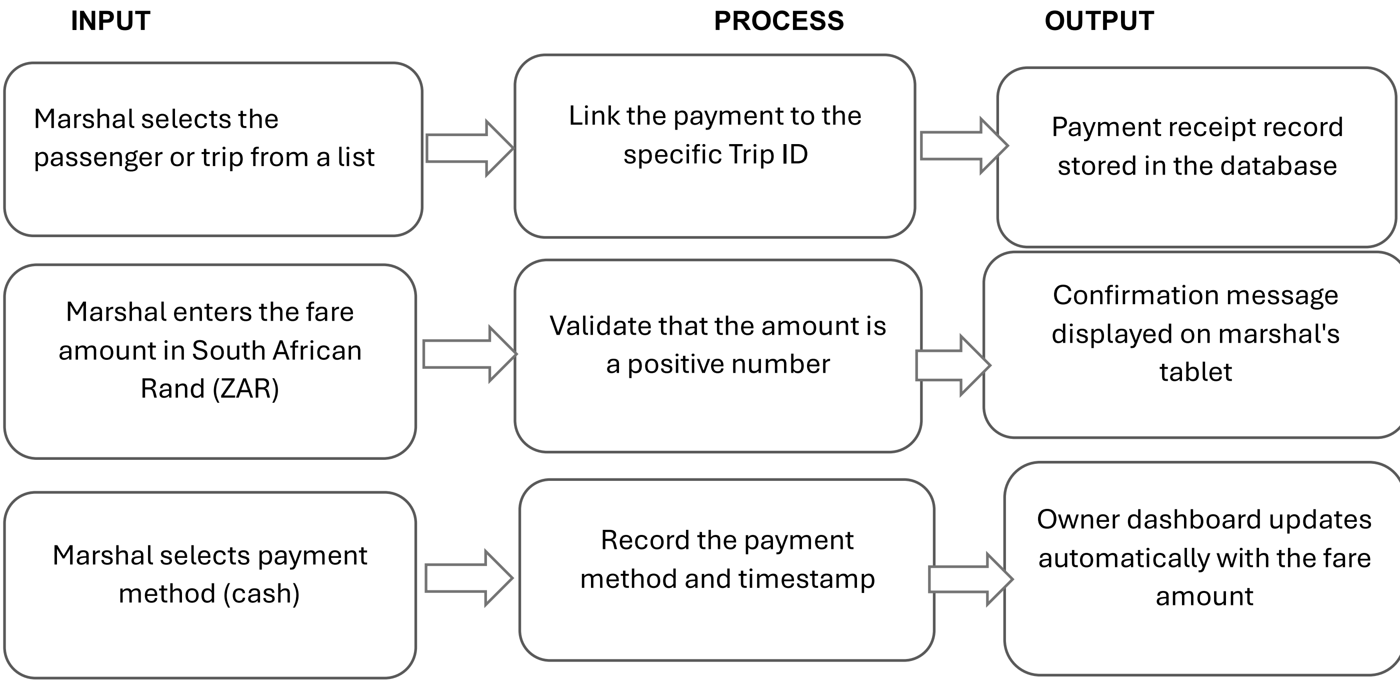
INPUT	PROCESS	OUTPUT
Passenger selects a starting taxi rank from a list	Query the database for all active taxis currently at that rank	List of available destinations (routes) from that rank
Passenger enters their current location	Calculate the next departure time for each route based on queue positions	Estimated wait time displayed for each route
Passenger taps "Check Availability" button	Display only routes that have at least one taxi available	Real-time taxi availability status (e.g., "Available now" or "15 min wait")

FR-05: Record Fare Collected from Passenger

User: Taxi Marshal

Problem Solved (from Phase 1): Manual fare recording leads to errors and makes it difficult for owners to track revenue.

Description: The system shall allow the taxi marshal to record the fare amount collected from each passenger. This fare is then linked to the specific trip and becomes visible to the taxi owner on their dashboard.

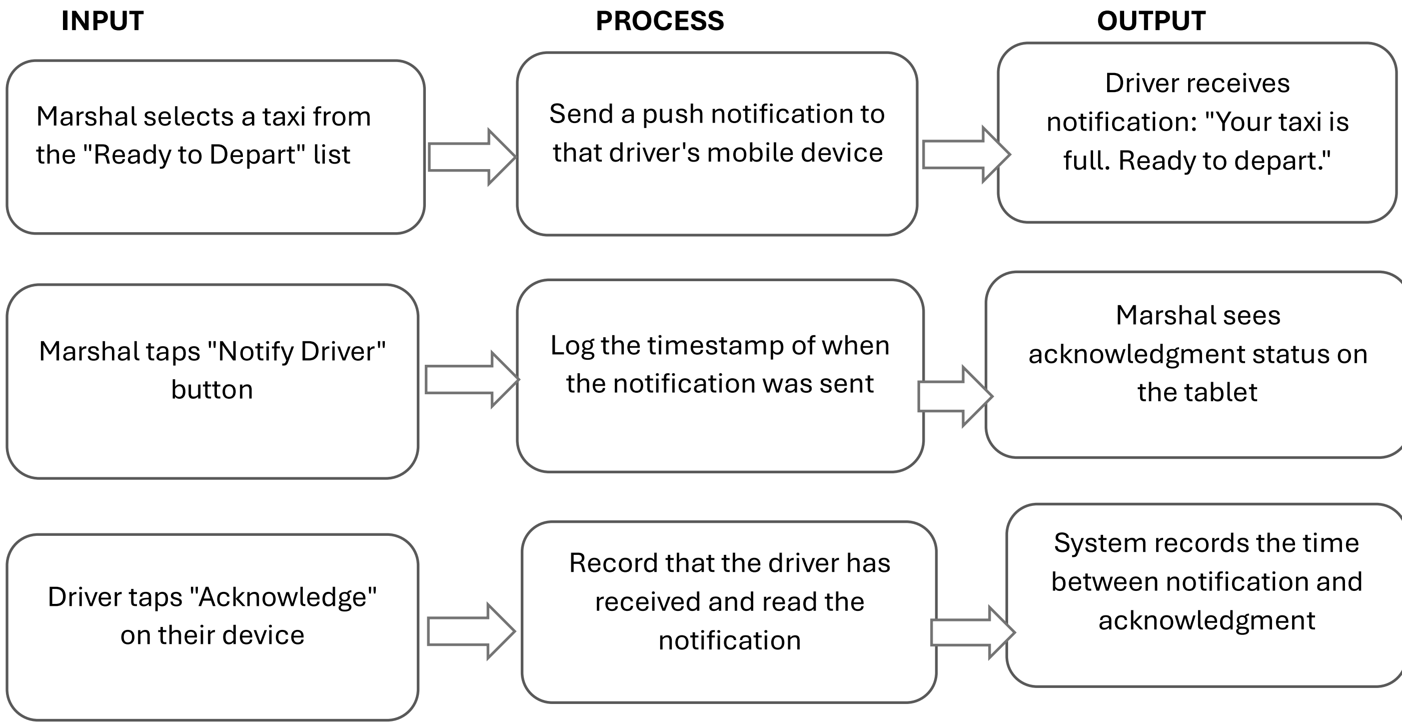


FR-06: Send Departure Notification from Marshal to Driver

User: Taxi Marshal

Problem Solved (from Phase 1): Communication between drivers, passengers, and taxi marshals is limited, resulting in inefficient taxi departures.

Description: The system shall allow the taxi marshal to send a notification to a specific driver when their taxi is full and ready to depart. The driver must acknowledge the notification, and the marshal can see that acknowledgment.



4. Non-functional Requirements:

- While functional requirements dictate what the system does, non-functional requirements define the performance level of those capabilities. Since the E-RANK system will be deployed in a challenging environment—characterised by unreliable connectivity, exposure to sunlight, and varying levels of technological literacy—these non-functional requirements are crucial for proper functionality.

All non-functional requirements listed include quantifiable goals. This provides a way to measure whether each requirement has been fulfilled. For example, instead of a vague statement like "the system should be fast," a precise timeframe of three seconds is specified for QR scanning. Instead of simply saying "the system should be secure," the AES-256 encryption standard is prescribed for passenger ID information.

The non-functional requirements are organised into seven categories: **Reliability, Availability, Security, Performance, Usability, Maintainability, and Portability.**

4.1 Reliability Requirements:

Reliability concerns the system's ability to operate without errors or failures. For E-RANK, high reliability is necessary because any queue data corruption could re-introduce disputes.

- The system must maintain the taxi queue with 99.9 percent accuracy, meaning no more than one queue entry in 1,000 may be lost or corrupted.
- There must be zero data loss for passenger records. The system must auto-save entered data every five seconds to prevent loss due to power or network issues.
- The system must match each passenger to the correct taxi and driver with 99.9 percent accuracy, preventing the assignment errors common in manual systems.

4.2 Availability Requirements:

Availability is the percentage of time the system is operational. Since taxi ranks operate during specific hours, 24/7 availability is not required.

- Core functions (queue management and passenger registration) must be available for 98 percent of operating hours, defined as 5:00 AM to 9:00 PM daily. This allows approximately 20 minutes of downtime per day for maintenance.
- The system must include a fallback mode for network outages. When internet connectivity is lost, the system must display the last-known queue list and continue functioning locally. When connectivity is restored, the system must automatically synchronise local data with the cloud database.

4.3 Security Requirements

Security requirements protect the system and its users from unauthorised access, data breaches, and privacy violations. E-RANK handles sensitive passenger information, making security a high priority.

- All passenger personal information (ID numbers, names, trip histories) must be encrypted using the AES-256 standard both in storage and during network transmission.
- For authentication, different user types require different security levels:
  - **Taxi owners:** Multi-factor authentication consisting of a password plus a one-time SMS code sent to their registered phone number.
  - **Taxi marshals:** Personal identification number (PIN) combined with a biometric factor, such as a fingerprint scan on the tablet device.
  - **Drivers and passengers:** Simpler authentication methods appropriate to their roles.

4.4 Performance Requirements

Performance requirements specify response times, throughput, and resource utilisation. Poor performance would frustrate users in the congested taxi rank environment and could lead to system rejection.

- QR code scanning and queue registration must complete within **three seconds** from scan to confirmation message on the driver's device.
- The owner dashboard, displaying up to 50 taxis, must fully load within **five seconds** when accessed over a standard 4G mobile network connection.
- The system must support at least **100 concurrent users** (marshals, drivers, and owners) without any degradation in response time.

4.5 Usability Requirements

Usability requirements ensure the system is easy to learn and use, particularly for users who may not be comfortable with technology.

- The marshal's tablet interface must allow queue management within a maximum of **two screen taps** (e.g., from home screen to "View Queue" to "Call Next Taxi").
- All touch targets on the tablet must be at least **44 by 44 pixels**, the recommended minimum for touch screens to accommodate different finger sizes.

- The tablet screen must be readable in direct sunlight, requiring a minimum brightness of **500 nits** and a text-to-background colour contrast ratio of at least **4.5:1**.

4.6 Maintainability Requirements

Maintainability is the extent to which the system can be modified to fix bugs, add features, or adapt to changing requirements.

- The E-RANK system must be built using a **modular architecture**, separating the backend server, marshal tablet application, driver mobile application, and owner web dashboard into distinct modules.
- Under this architecture, a developer should be able to modify one module (e.g., queue management) without disrupting other modules (e.g., passenger registration or owner dashboard).
- Modularity also allows different team members to work on different components simultaneously.

4.7 Portability Requirements

Portability requirements specify the system's ability to run on different hardware, operating systems, or environments. E-RANK must run on various devices because different taxi ranks may use different tablets, and drivers will have different smartphone models.

- The marshal tablet application must run on **Android 11+** and **iPadOS 15+**.
- The driver mobile application must run on **Android 9+** (smartphones) and **iOS 14+** (iPhones).
- The backend server must be deployable to major cloud platforms—**Amazon Web Services, Microsoft Azure, and Google Cloud Platform**—without requiring code changes. This ensures cost-effective hosting and avoids vendor lock-in.

Category	Measurable Target	Description
Reliability	99.9% accuracy	Queue order maintained with 99.9% accuracy. No passenger record loss.
Reliability	0% data loss	Auto-save every 5 seconds. Cloud backup of all records.
Availability	98% uptime	Core functions available 98% of operating hours (5 AM - 9 PM).
Availability	Fallback mode	During network outage, display last-known queue. Auto-sync when connection returns.

Category	Measurable Target	Description
Security	AES-256 encryption	Passenger ID numbers and personal details encrypted in storage and transmission.
Security	Multi-factor authentication	Owners: password + SMS code. Marshals: PIN + fingerprint.
Performance	< 3 seconds	QR code scanning to confirmation completes within 3 seconds.
Performance	< 5 seconds	Owner dashboard with 50 taxis loads within 5 seconds on 4G.
Performance	100+ concurrent users	System supports 100+ simultaneous users without slowdown.
Usability	2-tap maximum	Queue management requires maximum 2 screen taps.
Usability	44x44 pixel targets	All touch targets minimum 44x44 pixels. Screen readable in sunlight (500 nits).
Maintainability	Modular code	Backend, marshal app, driver app, owner dashboard are separate modules.
Portability	Cross-platform	Marshal tablet: Android 11+ and iPadOS 15+. Driver app: Android 9+ and iOS 14+.
Portability	Cloud deployment	Deployable to AWS, Azure, or Google Cloud without code changes.

5. UML Diagrams

Figure 5.1: Use Case Diagram showing all key users:

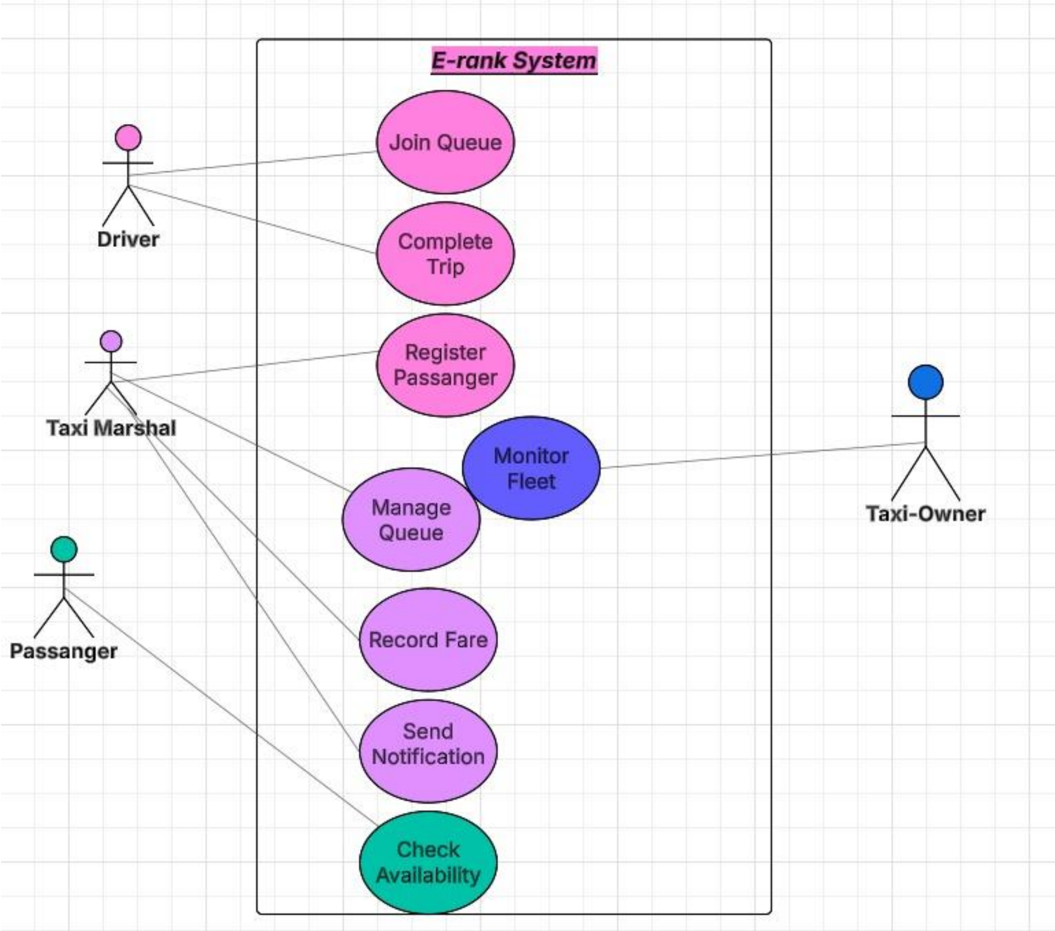


Figure 5.2: Context Diagram (System Boundaries):

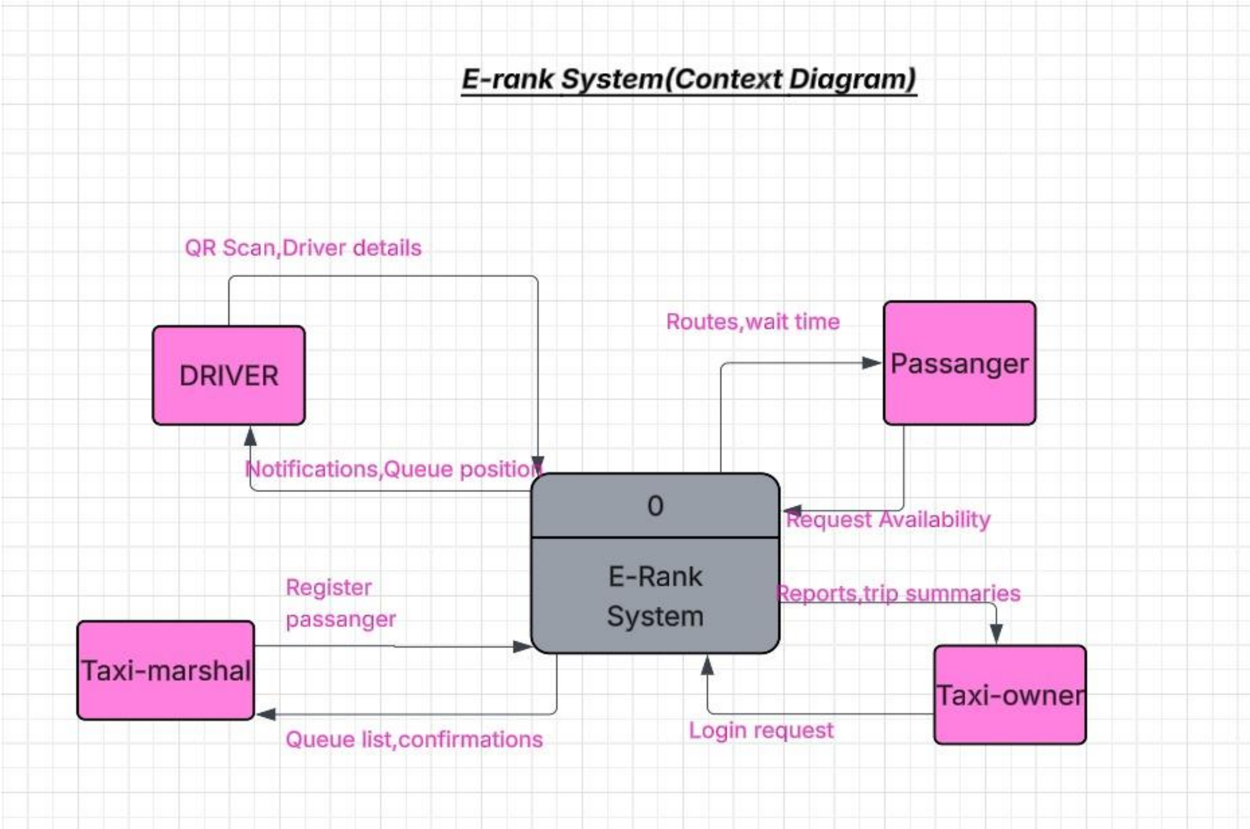


Figure 5.3: State Diagram – Driver State:

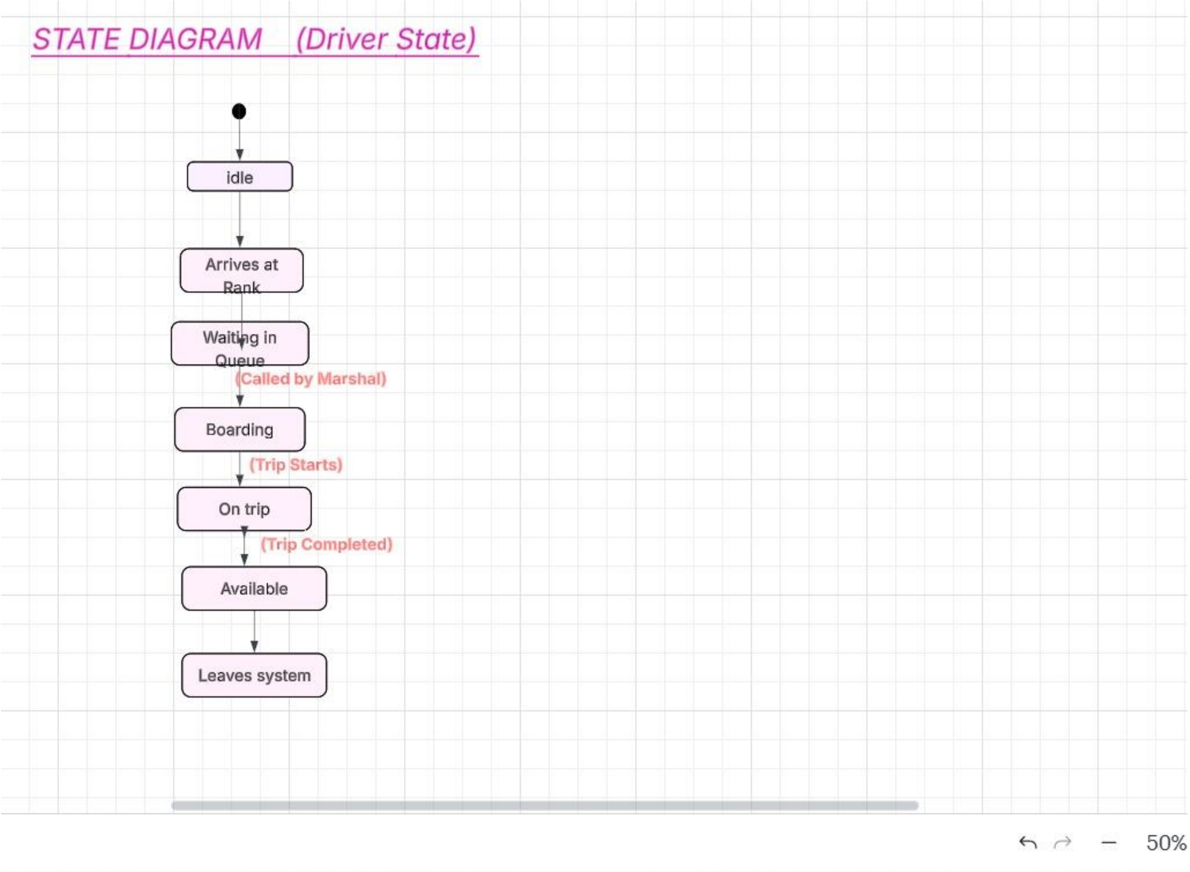


Figure 5.4: Sequence Diagram – Passenger Registration:

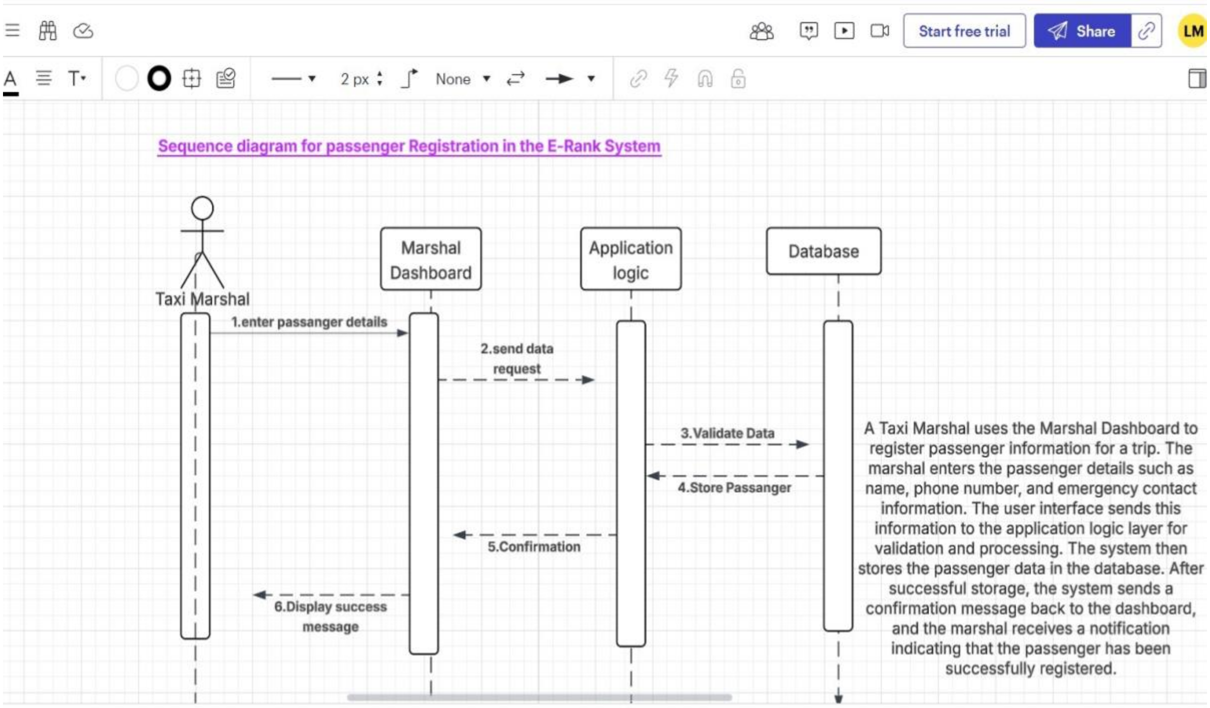
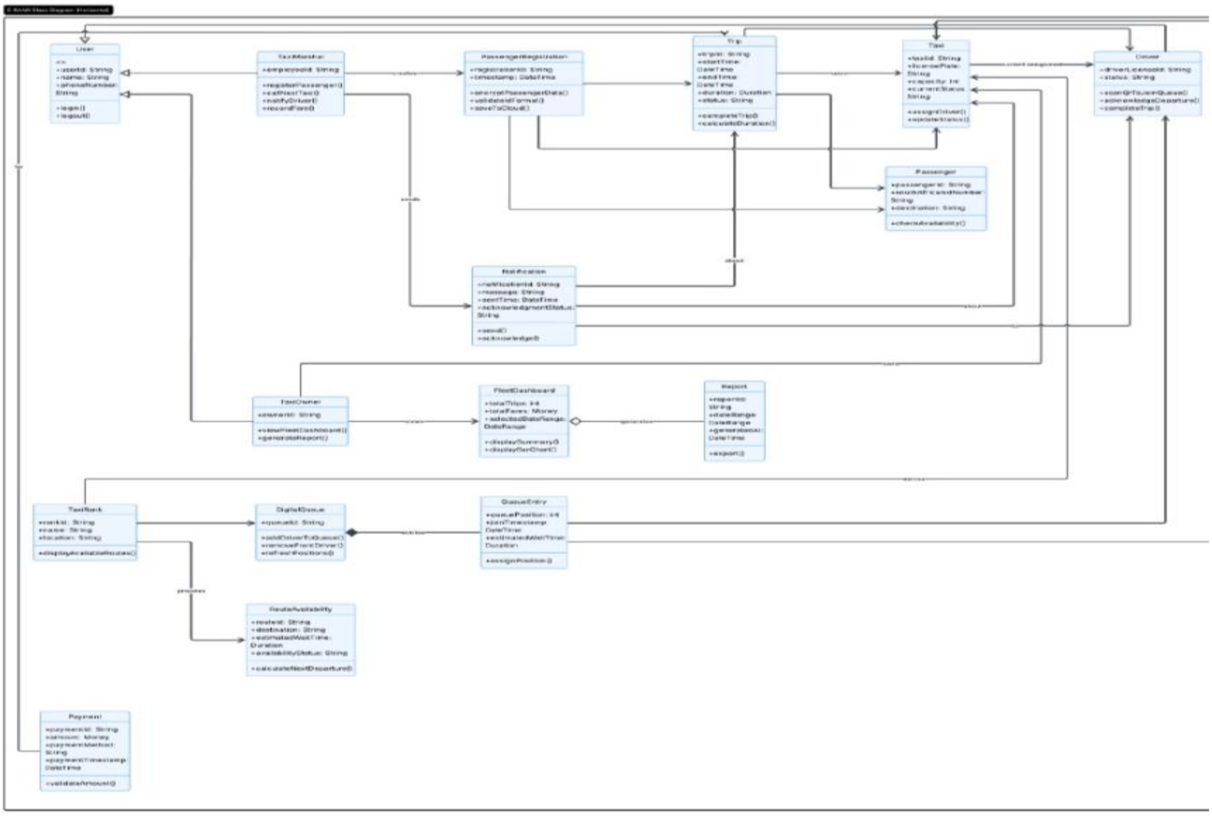


Figure 5.5: Class Diagram:



6. Project Management Artefacts

This section presents the key project management artefacts developed to ensure successful planning, execution, and monitoring of the E-RANK system. The Work Breakdown Structure (WBS) provides a detailed hierarchical decomposition of all tasks, deliverables, milestones, resources, dependencies, and risks. The Gantt Chart offers a clear visual timeline of the entire project schedule across the four phases.

Figure 6.1: Work Breakdown Structure:

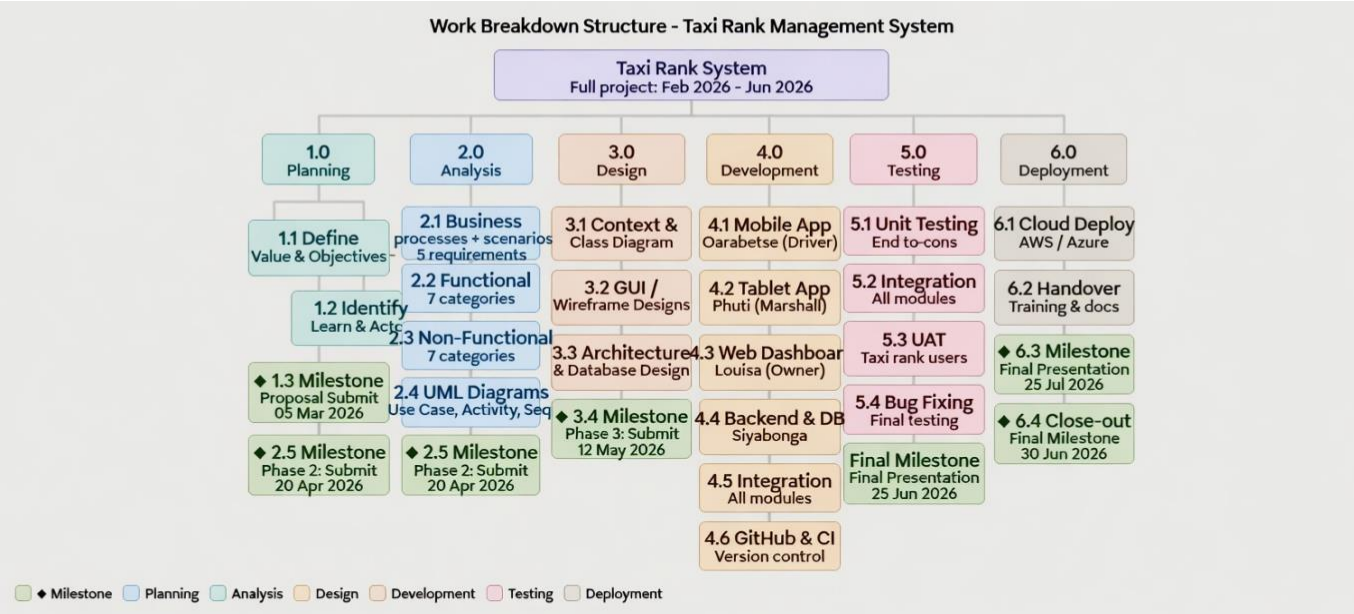


Figure 6.2: Gantt Chart – Project Timeline: